

Priyansh Gadia

ML Engineer | Quantitative Research | Full-Stack AI Systems

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SUMMARY

Dual-degree engineer (IIT Guwahati 9.25 GPA | DJSCE 8.23 GPA) shipping production AI and quantitative systems end-to-end - from research hypothesis to deployed containerized infrastructure. Architected Argus (institutional quant monorepo with strict DSR/PBO anti-overfitting discipline), CryptoGraph Analytics (live real-time platform forecasting 107 crypto assets), and clinical ML models validated across 50,920 ICU records. Combines deep mathematical rigor (PyTorch, GNNs, state spaces) with enterprise engineering (FastAPI, Docker, Next.js 14) to build robust, high-conviction systems.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, SQLite), C, C++, Java, R, MATLAB, TypeScript, JavaScript
ML & Deep Learning: PyTorch, LightGBM, XGBoost, scikit-learn, ST-GCN, LSTM, NeuralProphet, SHAP, SciPy
Quantitative & Risk: Mamba-3, GAT, Path Signatures, HRP, Kelly Sizing, Meucci Entropy Pooling, DSR/PBO, CPCV
Systems & Cloud: FastAPI, Next.js 14, Docker, Redis, Streamlit, PostgreSQL, WebSockets, Alembic, Prometheus
Platforms & Data: Oracle Cloud Infrastructure (Certified), Binance API, MIMIC-IV, Groq LLM API, ReportLab

PROJECTS

Argus - Institutional Quantitative & RegTech Operating System *[Ongoing]* [\[GitHub\]](#)

Python 3.12, PyTorch, LightGBM, Streamlit, ReportLab, Alembic, Mypy strict, 73 unit tests

- Architecting an 11-package monorepo for quantitative asset management: feature store (fractional differencing, rough volatility, path signatures), alpha models (RSMG-Net: Mamba-3 + GAT; RSMG-Lite: GBDT), and portfolio risk budgeting (HRP, meta-gated Kelly, Meucci entropy pooling).
- Enforced institutional anti-overfitting protocol (CSCV, Deflated Sharpe Ratio with N-trial penalty, PBO); built RegTech engine with SEBI Master Circular 2026 compliance, SHA-256 audit ledger, and automated equity research notes under Mypy strict typing.

CryptoGraph Analytics - Ensemble Forecasting & Real-Time Trading Platform *[Live]* [\[GitHub\]](#)

Python, PyTorch, FastAPI, Next.js 14, Docker, Redis, PostgreSQL (Supabase)

- Architected a full-stack platform ingesting Binance WebSocket feeds, computing multi-modal indicators across 107 crypto assets streamed to a live Next.js 14 dashboard with interactive Recharts visualizers.
- Engineered an ensemble pipeline (ST-GCN multi-relational graphs x LSTM x NeuralProphet) gated by Sharpe ratio thresholds (> 1.5), integrated with a Mixture-of-Agents LLM explainability swarm and Redis/Docker backend.

Respiratory Support Optimization - Biomedical ML Research [\[GitHub\]](#)

Python, SciPy, scikit-learn, pandas, NumPy

- Led a 3-phase study detecting clinically invisible ICU pressure transients; processed 200 Hz physiological waveforms validated against 50,920 MIMIC-IV patient records.
- Built the first patient-ventilator asynchrony detector anchored in esophageal pressure ground truth ($PL = P_{aw} - P_{es}$) using bedside-only allowlist features under LOPO cross-validation across 1,405 synthetic runs.

SELECTED TECHNICAL PROJECTS

- **PhysioNet Challenge 2026** [\[GitHub\]](#) - *LightGBM, XGBoost, Docker*: Predicting cognitive impairment across 6,600 PSG records (3 hospital sites); extracted 120+ sleep/EEG/HRV features with dual-layer age residualization ($f_{age_z} + \text{post-hoc } \gamma$) in a Dockerized pipeline.
- **Intelligent Document Processing Engine** [\[GitHub\]](#) - *FastAPI, XGBoost, LayoutLMv3*: Enterprise doc intelligence with 1:1 OCI Cloud Adapter pattern, 4-tier IntelligentRouter, and SHA-256 prediction cache for $O(1)$ response in <150 MB RAM.
- **GPT-2 From Scratch** [\[GitHub\]](#) - *Python, PyTorch*: Autoregressive transformer (multi-head attention, positional embeddings, causal masking) implemented end-to-end in PyTorch.

EDUCATION

Indian Institute of Technology Guwahati	Oct 2023 - Aug 2027
B.Sc. (Hons) Data Science & Artificial Intelligence GPA: 9.25 / 10.0	
Dwarkanadas J. Sanghvi College of Engineering	Aug 2023 - Aug 2027
B.Tech (Hons) Mechanical Engineering & Robotics GPA: 8.23 / 10.0	

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Data Science Professional - Oracle University	Oct 2025
CITI Human Subjects Research - MIT Affiliate	Jul 2025
Data Mining of Clinical Databases (CDSS) - University of Glasgow	Jul 2025
University of Michigan - Programming for Everybody (Python) - edX ID: cb80becb7b7044fe85f6a17f2282e12e	2021